

Capstone One-Pager

Backend Engineering Launchpad

SELECTED TOPIC

Draftly: Gmail AI Reply Agent

Problem

PROBLEM STATEMENT

Professionals spend time writing repetitive email replies such as confirmations, follow-ups, and acknowledgements. The system should generate smart reply drafts, preserve thread context, and keep the user in full control before anything is sent.

Approach Step-by-step

STEP-BY-STEP APPROACH

- Design the backend flow**
Define APIs for Gmail connect, inbox fetch, draft generation, review, approval, send, and logs.
- Integrate Gmail securely**
Use OAuth2 for access, fetch unread/recent messages, preserve thread IDs, headers, and metadata for replies.
- Generate drafts with Gemini**
Pass thread context, tone preference, signature, and recent sent emails so the draft matches the user's style.
- Add human review before send**
Store generated drafts and allow approve, edit, or reject actions so AI assistance stays supervised.
- Make sending reliable**
Use retries, idempotency keys, encrypted credentials, and logs for safe sending, debugging, and operational visibility.

HIGH-LEVEL DESIGN

FastAPI APIs manage auth, inbox, drafts, review, send, and logs. Gmail handles mailbox access, Gemini generates replies, and SQLite stores users, preferences, drafts, send attempts, and audit history.

Outcome

Final goal:
an extensible Gmail AI reply backend
with human approval.

Selected topic: Draftly | Stack: FastAPI, Gmail API, Gemini, SQLite